Cell Matching × Segmentation

CS-GY 6643 · Computer Vision · NYU Tandon

Project 2, Question 5 · Fall 2026

Team size	Up to 2 participants
Submissions	5 per day
Metric	Custom (see Section 5)
Competition closes	Thursday 8 October 2026, 11:59 pm
Report and code due	Sunday 11 October 2026, 11:59 pm

Contents

1 Overview **2**

2 Description **2**

3 Problem Domain **2**

4 The Challenge **4**

5 Evaluation **5**

6 Dataset Description **7**

1. Overview

This competition focuses on linking the same neurons across two different imaging settings.

In-vivo imaging records neurons in the living mouse and captures their activity. Ex-vivo imaging is performed after the tissue is removed and processed, allowing researchers to identify properties such as cell type and gene expression.

The challenge is that the same neuron can look very different across these two imaging methods. To connect the functional information from in-vivo imaging with the biological information from ex-vivo imaging, the corresponding cells must be identified across both images.

For each paired in-vivo and ex-vivo image, your task is to:

- segment all cells in the in-vivo image,
- segment all cells in the ex-vivo image,
- match corresponding cells across the two images.

The training data includes cell masks and verified cell matches.

2. Description

The dataset is organized into **regions**. Each region contains one in-vivo image and one ex-vivo image from the same mouse:

- `invivo.tif`
- `exvivo.tif`

Both are 16-bit grayscale TIFF images. Image sizes vary across regions, so always read the actual image dimensions rather than assuming a fixed size.

No alignment between the in-vivo and ex-vivo images is provided.

Another important challenge is that **most cells do not have a verified match**. The training data contains **1,139 verified cell pairs across 20,409 cells**. A cell that does not appear in the correspondence list should be treated as **unverified**, not necessarily as having no match.

Segmentation and matching are evaluated together. A predicted correspondence is only counted as correct if the corresponding cell masks are also correct.

Participants may compete individually or in teams of up to **2 members**.

3. Problem Domain

This task is known as **cross-modal cell registration**.

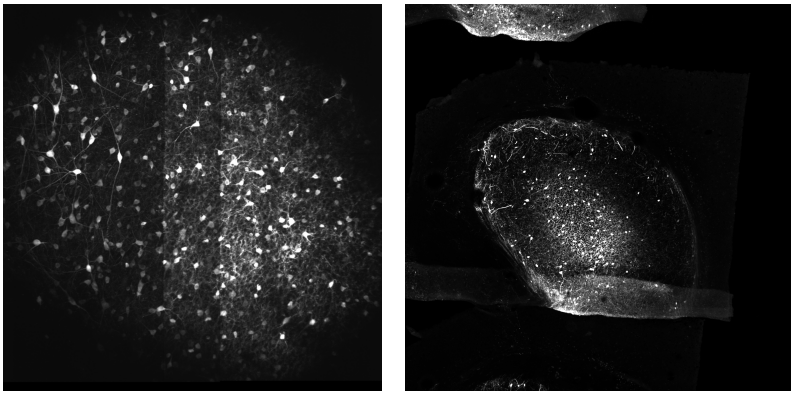


Figure 1: One region: the in-vivo image (left) and the ex-vivo image (right) from the same mouse, shown here at equal display height. In pixels the ex-vivo image is several times larger. Training region `subject_db6b8b__region_65121e`.

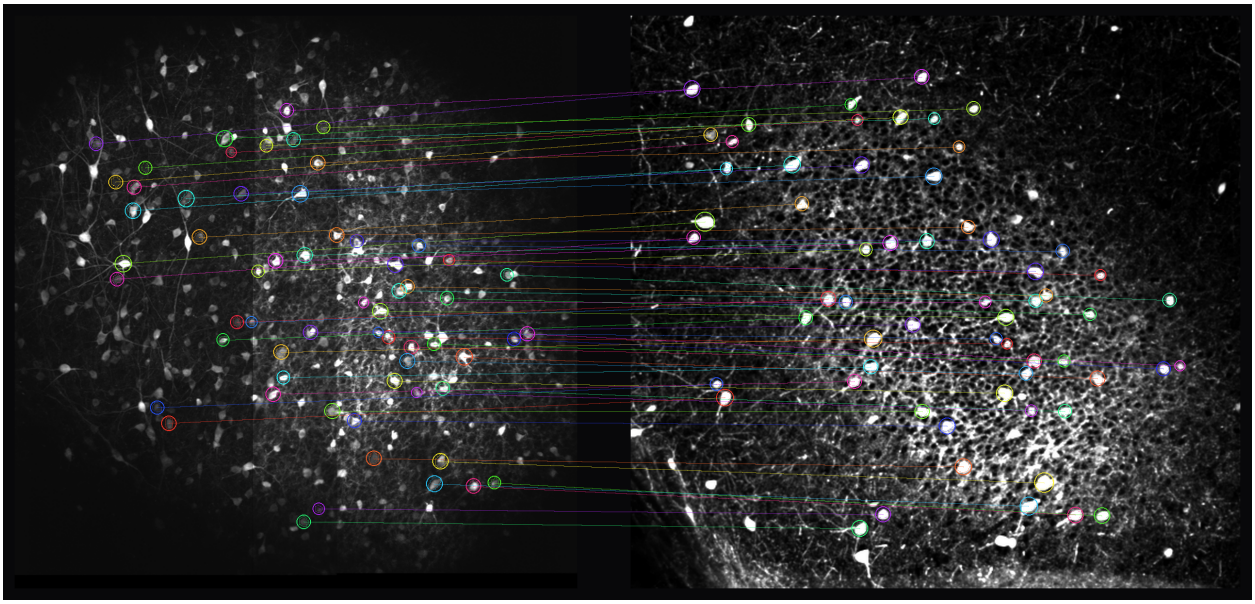


Figure 2: The verified correspondences for the region above: 61 pairs among 358 in-vivo and 163 ex-vivo cells. Each pair is drawn in its own colour, with a thin outline on the annotated mask. The ex-vivo panel is cropped to the area containing the matched cells.

The goal is to connect functional information from in-vivo imaging with biological information from ex-vivo imaging by identifying the same neurons across both modalities.

This is difficult because the two images are captured under very different conditions. Tissue deformation, different optics, contrast changes, sectioning, and missing cells can all change how the same neuron appears.

Individual cell appearance is often not enough for reliable matching. Many neurons look similar, so the **relative arrangement and geometry of cells** becomes an important source of information.

This problem is closely related to image registration and point-set alignment.

4. The Challenge

For **every region** in the test set:

1. **Segment the in-vivo image.** Report one instance mask per cell, each with its own id.
2. **Segment the ex-vivo image** in the same way; this modality is more challenging.
3. **Match cells across the two images.** Report a list of pairs `[invivo_id, exvivo_id]`, each asserting that the two masks are the same neuron.

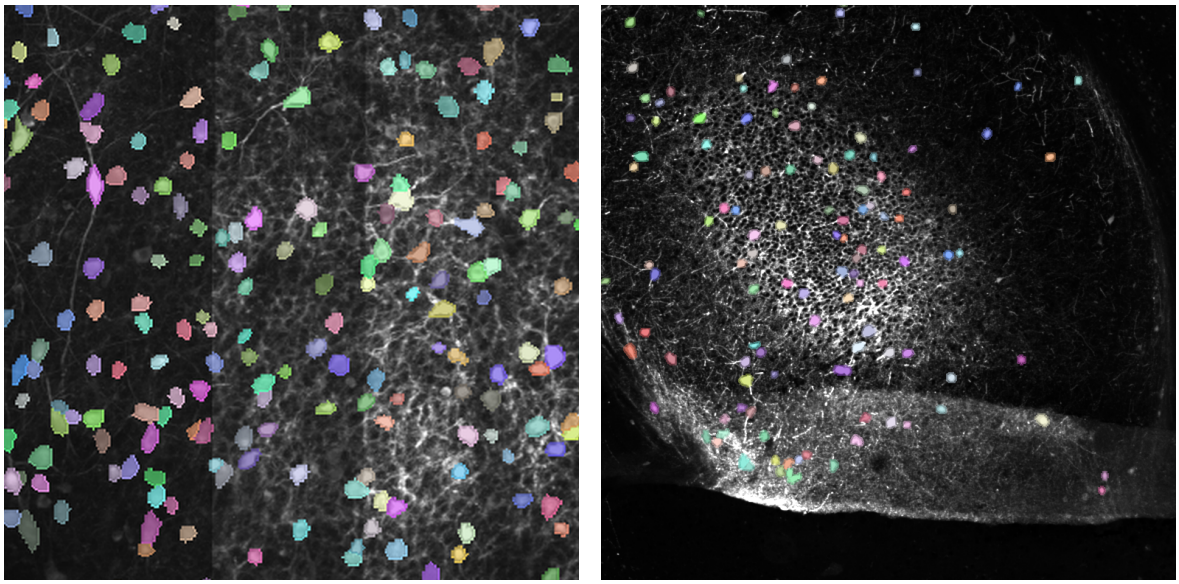


Figure 3: Annotated instance masks, magnified, in-vivo (left) and ex-vivo (right). Each cell carries its own mask and its own identifier; adjacent cells are separate instances. Taken from the same training region.

Technical challenges

1. **Intensity alone does not identify a cell body.** In the ex-vivo images, neuronal processes reach intensities similar to cell bodies. Spatial and morphological information must be used alongside intensity.

2. **The two images are at different scales.** Cells occupy a similar number of pixels in both modalities: across the training set the median cell area is 86 pixels in vivo and 71 pixels ex vivo. The ex-vivo frame, however, is several times larger, so a cell is a much smaller fraction of it. A detector tuned to one modality may not generalize well to the other.

3. **Each cell needs its own mask, not a cell/background map.** Labelling every pixel as cell or background is not enough. Touching cells must be separated into individual instances before matching, each with its own mask and id.

Rules

- **No hardcoding.** Your code must run unchanged on regions it has never seen. Your submitted code must reproduce your leaderboard submission; scores that cannot be reproduced from the submitted code will not count.
- A submission that reuses a cell id across two pairs is rejected.

5. Evaluation

Segmentation and matching contribute equally to the final score. In-vivo PQ and ex-vivo PQ each contribute 25%, and matching F1 contributes 50%.

$$S \;=\; 0.50 \cdot \frac{\mathrm{PQ}_{\text{in-vivo}} + \mathrm{PQ}_{\text{ex-vivo}}}{2} \;+\; 0.50 \cdot \mathrm{F1}_{\text{match}}$$

Term	Weight	Measures
$\mathrm{PQ}_{\text{in-vivo}}$	25%	instance masks in the in-vivo image
$\mathrm{PQ}_{\text{ex-vivo}}$	25%	instance masks in the ex-vivo image
$\mathrm{F1}_{\text{match}}$	50%	the in-vivo $\leftrightarrow$ ex-vivo correspondences

5.1 Mask Matching with IoU

Both terms depend on a single matching rule. A predicted mask P is compared to a ground-truth mask G by Intersection over Union, where $|X|$ is the pixel count of X:

$$\mathrm{IoU}(P, G) \;=\; \frac{|P \cap G|}{|P \cup G|}$$

A predicted mask is a **true positive** only if its IoU with a ground-truth mask exceeds **0.75**. Each ground-truth cell may be claimed once. Predictions matching no ground-truth cell are **false positives**; ground-truth cells claimed by no prediction are **false negatives**.

5.2 Segmentation: Panoptic Quality

Panoptic Quality is computed separately for each modality and each region, then averaged over regions.

$$\mathrm{PQ} \;=\; \frac{\sum_{(p,g)\in \mathrm{TP}} \mathrm{IoU}(p,g)}{|\mathrm{TP}| + \frac{1}{2}|\mathrm{FP}| + \frac{1}{2}|\mathrm{FN}|}$$

5.3 Matching: F1

A predicted pair $[i, j]$ is **correct** only if masks i and j are both true positives *and* the two ground-truth cells they matched are a verified pair. Precision and recall are pooled over all test regions.

$$\text{precision} = \frac{\text{correct pairs}}{\text{predicted pairs}}, \qquad \text{recall} = \frac{\text{correct pairs}}{\text{verified pairs}}, \qquad \mathrm{F1} = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

5.4 Submission File

One row per test region, 29 rows plus a header. Every `sample_id` listed in `sample_submission.csv` must appear exactly once. The header is:

```
sample_id,invivo_instances,exvivo_instances,match_pairs
```

- `sample_id`: the region identifier, `<subject>__<region>`, e.g. `subject_78b6a7__region_33daed`. Copy it verbatim from the sample submission.
- `invivo_instances`: a JSON object mapping your cell id to that cell's RLE mask, `{cell_id: rle}`. One entry per cell detected in the in-vivo image.
- `exvivo_instances`: the same, for the ex-vivo image.
- `match_pairs`: a JSON list of two-element lists, `[[invivo_id, exvivo_id], …]`. Every id used must be a key defined in the same row, and no id may appear more than once.

5.5 Deadlines

Due	What	Where
Thu 8 Oct 2026, 11:59 pm	Final Kaggle submission	Kaggle
Sun 11 Oct 2026, 11:59 pm	Report and runnable code	Brightspace

On Kaggle manually select your final submission. Only the selected submission will be scored on the private leaderboard. All other submissions will not be considered.

5.6 Scoring

The Kaggle component is worth **65 points**, made up as follows:

Component	Points
Report and code	5
Kaggle private leaderboard	40
Beating the baseline on the private leaderboard	15
Reaching #1 at any point (one-time)	5
Total	**65**

- **Report and code (5 points).** Your Question 5 report, written from the report template, together with the runnable code that produces your submission. State what your method does and why, not how the code is organised.
- **Private leaderboard (40 points).** Your final standing on the private leaderboard, scored with the metric in Section 5.
- **Beating the private leaderboard's baseline (15 points).** Awarded for finishing above the baseline entry on the private leaderboard.
- **Reaching #1 (5 points).** Each student whose team reaches **#1** and beats the private leaderboard baseline at any point receives a one-time **5-point bonus**. This includes the final private-leaderboard winner. The bonus is earned at most once per team per competition.

6. Dataset Description

The dataset contains training data, hidden test data, and a sample submission file.

6.1 Training Data

The training set contains **47 regions from 3 mice**.

Each region contains:

```
invivo.tif
exvivo.tif
```

The training set also includes:

```
train_ground_truth.csv
```

which contains the segmentation masks and matched cell pairs for all training regions.

Overall, the training set contains:

- **14,893 in-vivo cells**

- **5,516 ex-vivo cells**
- **1,139 matched cell pairs**

6.2 Hidden Test Data

The hidden test set contains **29 regions from 2 mice**.

Each region contains only:

```
invivo.tif
exvivo.tif
```

The ground truth is hidden and used for leaderboard evaluation.

6.3 Folder Structure

```
training/
|-- train_ground_truth.csv
|-- <subject>/
|   `-- <region>/
|       |-- invivo.tif
|       `-- exvivo.tif

hidden_test/
|-- <subject>/
|   `-- <region>/
|       |-- invivo.tif
|       `-- exvivo.tif

sample_submission.csv
```

Each region is identified using:

```
sample_id = <subject>__<region>
```

Example:

```
subject_5d294c__region_2b8b0c
```

6.4 `train_ground_truth.csv`

The file contains **47 rows**, one for each training region.

It has four columns:

Column	Description
`sample_id`	Region identifier
`invivo_instances`	In-vivo cell masks
`exvivo_instances`	Ex-vivo cell masks
`match_pairs`	Matched in-vivo and ex-vivo cell IDs

The instance masks are stored as JSON objects using RLE.

The matched cells are stored as:

```
[
  ["IV_000031", "EX_000014"],
  ["IV_000052", "EX_000021"]
]
```

6.5 `sample_submission.csv`

`sample_submission.csv` contains one row for every hidden test region and shows the exact format required for submission.

The `sample_id` values are already filled in.

The prediction columns are initially empty:

```
invivo_instances = {}
exvivo_instances = {}
match_pairs      = []
```

Use this file as the base format for your final submission.